NARAYANA SCHOOLS - WEST BENGAL

PRE-BOARD EXAMINATION

SUBJECT: SCIENCE (086) FULL MARKS: 80 TIME:3.00Hrs. DATE: 25.11.24

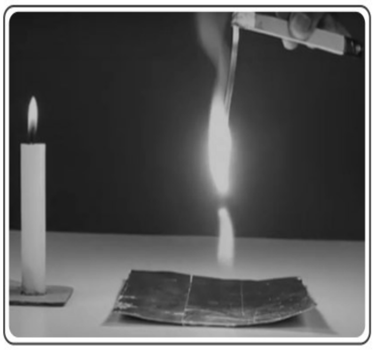
CLASS: X

GENERAL INSTRUCTIONS

Read the following instructions very carefully and strictly follow them:

1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

Q. No.	QUESTIONS	Marks
	SECTION A *This section comprises multiple choice questions (MCQs)/AR of 1 mark each.* **[20×1=20]**	
1.	Some types of chemical reactions are listed below. Which two of the following chemical reactions are of the SAME type? P) $AgNO_3 + NaCl \text{ ---> } AgCl + NaNO_3$ Q) $Mg + 2\ HCl \text{ ---> } MgCl_2 + H_2$ R) $CH_4 + 2\ O_2 \text{ ---> } CO_2 + 2\ H_2O$ S) $2\ KOH + H_2SO_4 \text{ ---> } K_2SO_4 + H_2O$ (a) P and Q (b) Q and R (c) R and S (d) P and S	1
2.	While burning firecrackers, we experience fire with multiple colours. One of the ways to achieve this is by mixing different metal powders while manufacturing the crackers. For instance, copper burns with a bright green colour and magnesium with a bright white colour. Let us dig deeper into this simple but enjoyable experiment with burning magnesium which can be easily performed in a lab or even at home. The lab that experimented with the oxides above explored the burning of magnesium. The group burnt a 10 cm long magnesium strip by placing it in a candle flame and recorded the time taken for the strip to catch fire as well as to burn fully. Later the strip was burnt by exposing it to another magnesium strip, which was already burning, and the two times values were recorded. According to you, which of the following is the root cause of the behaviour seen in the first row of the observations above? Note that we often have one cause leading to an effect which causes another effect and so on. Root cause refers to the first cause.	1



	Exposed to a candle flame			Exposed to a burning magnesium strip		
	Trial 1	Trial 2	Trial 3	Trial 1	Trial 2	Trial 3
Time to catch fire (s)	14	12	15	2	1	2
Time to burn full strip (s)	8	7	9	8	8	7

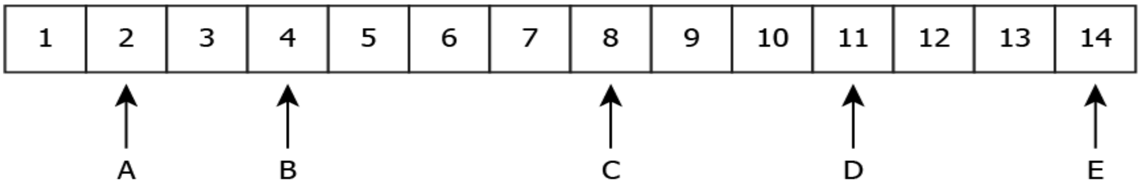
(a) The candle flame has less oxygen supply compared to the magnesium flame.
(b) The rate at which a candle flame transfers heat is less than the rate at which it is transferred from the magnesium flame.
(c) Temperature of the candle flame is lower than that of the magnesium flame.
(d) As the candle wick burns, it produces CO_2, which keeps extinguishing the magnesium and hence it takes longer to catch fire.

3. Water of crystallization is the fixed number of water molecules present in one formula unit of a salt. Find out which of the following is the correct match? 1

	Name of the salts		No. of water molecules as water of crystallisation
A.	Gypsum	1.	5
B.	Washing soda	2.	$\frac{1}{2}$
C.	Plaster of paris	3.	10
D.	Hydrated $CuSO_4$	4.	2

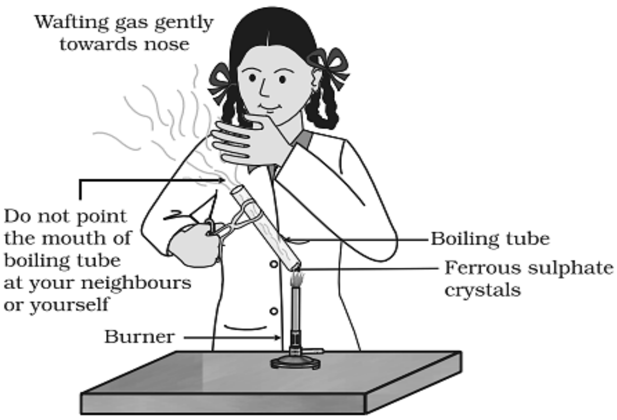
(a) D-1, A-2, B-3, C-4 (b) D-1, C-2, B-3, A-4
(c) B-1, C-2, A-3, D-4 (d) B-1, A-2, C-3, D-4

4. The image shows five solutions labelled on a pH scale. 1



Which classification is correct?
(a) Strongest Acid: B ; Strongest Base: E (b) Strongest Acid: A ; Strongest Base: E
(c) Strongest Acid: A ; Strongest Base: C (d) Strongest Acid: B ; Strongest Base: C

5. Reeta heated ferrous sulphate crystals, and observed the followings in chemistry laboratory. Find out which is/ are correct? 1



	I. Characteristic odour of burning sulphur. II. Solid ferric oxide is formed. III. O_2 and H_2O are formed. IV. Powdered FeS is formed choose the correct option. (a) II and III (b) III and IV (c) I and II (d) I and IV	
6.	The electronic configuration of three elements X, Y and Z are X- 2, 8, 7; Y- 2,8, 2; and Z - 2, 8 (a) Y and Z are metals (b) Y and X are non-metals (c) X is a non -metal and Y is a metal (d) Y is a non-metal and Z is a metal	1
7.	$KNO_3 + \boxed{Z\ (\text{yellowppt})} \xleftarrow{KI} \boxed{X} \xrightarrow{HEAT} \boxed{W} + NO_2 + O_2(g)$ In the above sequence of reactions, identify X, W and Z. (a) X= lead oxide, W= lead nitrate, Z= lead iodide (b) X= lead nitrate, W= lead oxide, Z= lead iodide (c) X= lead nitrate, W= lead iodide, Z= lead oxide (d) X= lead iodide, W= lead nitrate, Z= lead oxide	1
8.	The opening and closing of the stomatal pore depend upon: (a) Oxygen (b) Temperature (c) Water in the guard cells (d) Concentration of CO_2	1
9.	A plant gets rid of excess water through transpiration. Which is a method used by plants to get rid of solid waste products? (a) shortening of stem (b) dropping down of fruits (c) shedding of yellow leaves (d) expansion of roots into the soil	1
10.	How is food transported from phloem to the tissues according to plants' need? (a) Food is transported along with the water in the plant's body. (b) Food is transported in only one direction like water in the plant body through xylem. (c) Food is transported from a region with low concentration to higher concentration. (d) Food is transported from a region where it is produced to other parts of the plants.	1
11.	Which is the correct sequence of the components of a reflex arc? (a) Receptors → Muscles → Sensory neuron → Motor neuron → Spinal cord (b) Receptors → Motor neuron → Spinal cord → Sensory neuron → Muscle (c) Receptors → Spinal cord → Sensory neuron → Motor neuron → Muscle (d) Receptors → Sensory neuron → Spinal cord → Motor neuron → Muscle	1
12.	Coagulation of blood in a cut or wound is brought about by: (a) Plasma (b) Platelets (c) WBC (d) RBC	1
13.	When light enters the atmosphere, it strikes on extremely fine particles, which deflect the rays of light in all possible directions. This is due to - (a) reflection of light (b) atmospheric refraction (c) scattering of light (d) dispersion of light	1
14.	A complete circuit is left on for several minutes, causing the connecting copper wire to become hot. As the temperature of the wire increases, the electrical resistance of the wire (a) decreases. (b) remains the same. (c) increases. (d) increases for some time and then decreases.	1
15.	In pea plants, the pods may be inflated (I, dominant) or constricted (i, recessive). What proportion of the offspring would be expected to be inflated, if (Ii) is crossed with (ii)? (a) 25% (b) 50% (c) 75% (d) 100%	1

16.	In a food chain, 10,000 joules of energy are available to the producer. How much energy will be available to the secondary consumer to transfer it to the tertiary consumer? (a) 10 J (b) 100 J (c) 1000 J (d) 10,000 J	1

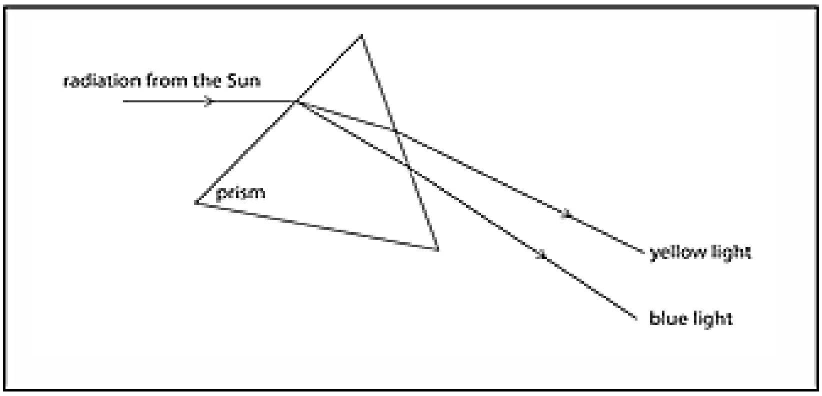
Read the assertion and the reason (Qs 17-20) given below and choose the correct option.
(a)Both the assertion and the reason are correct and the reason is the correct explanation for the assertion.
(b)Both the assertion and reason are correct but the reason is not the correct explanation for the assertion.
(c)The assertion is true but the reason is false.
(d)The assertion is false but the reason is true.

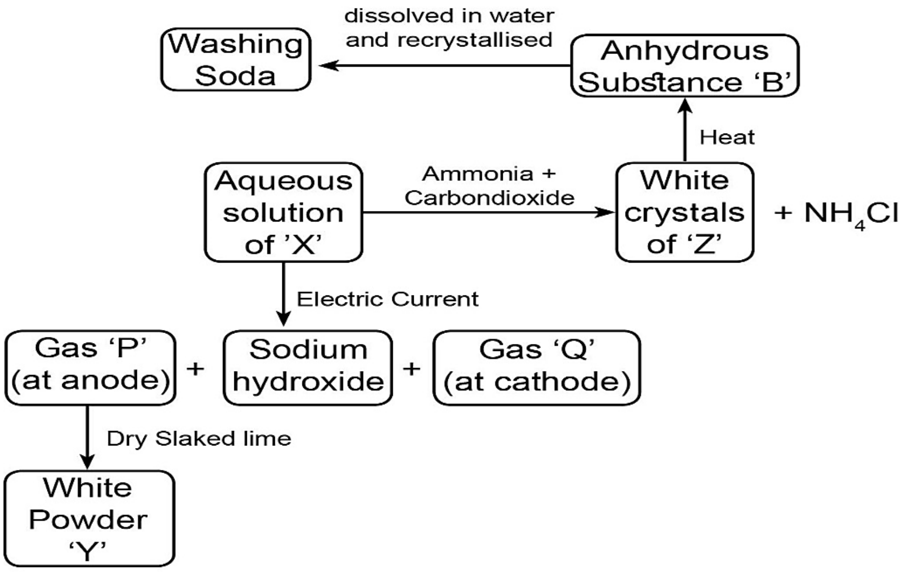
17.	**Assertion (A):** On adding dil. HCl to a test tube containing a substance 'X', a colourless gas is produced which gives a pop sound when a burning match stick is brought near it. **Reason (R):** In this reaction metal 'X' is displaced by Hydrogen.	1
18.	**Assertion(A):** Mendel selected the pea plant for his experiments. **Reason (R):** Pea plant is cross-pollinating and has unisexual flowers.	1
19.	**Assertion(A):** A compass needle is placed near a current carrying wire. The deflection of the compass needle decreases when the magnitude of the current in the wire is increased. **Reason (R)**: The strength of a magnetic field at a point near the conductor increases on increasing the current.	1
20.	**Assertion(A):** Polythene bags and plastic containers are non-biodegradable substances. **Reason(R):** They can be broken down by microorganisms in natural simple harmless substances.	1

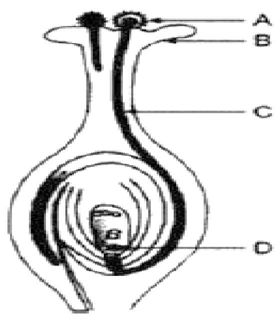
SECTION B

This section comprises very short answer (VSA) type questions of 2 marks each.

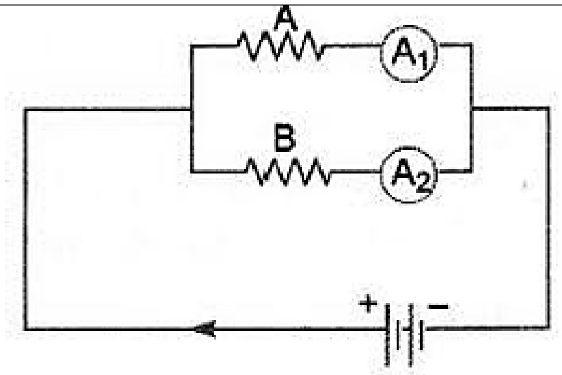
[6x2=12]

21.	State and explain the reason for the change in appearance observed when each of the following metals is exposed to atmospheric air for some time: (a) Iron (b) Copper	2
22.	What constitutes the central and peripheral nervous systems? How are the components of central nervous system protected?	2
23.	(a) What happens when the adrenaline is secreted into the blood? **OR** (b) State how concentration of auxin stimulates the cell to grow longer on the side of the shoot which is away from light?	2
24.	 State the phenomena observed in the above diagram. Explain with reference to the diagram, which of the two lights mentioned above will have the higher wavelength?	2
25.	How will you use two identical prisms so that a narrow beam of white light incident on one prism emerges out of the second prism as white light? Draw the diagram.	

	OR Imagine a current carrying circular loop of wire on the plane of your answer sheet. The magnetic field inside the loop is into the plane of the paper. i) What must be the direction of the current in the loop? ii) State the rule used here.	2
26.	Which gas shield the surface of earth from harmful radiation of the sun? Why these radiations are supposed to be harmful for us?	2
	SECTION C *This section comprises short answer (SA) type questions of **3** marks each.* **[7 x 3 = 21]**	
27.	A teacher asks her students to identify a metal, M. She gives them the following clues to help them. (P) Its oxide reacts with both HCl and NaOH. (Q) It does not react with hot or cold water but reacts with steam. (R) It can be extracted by electrolysis of its ore. (a) Identify the metal. (b) Write the chemical equations for the reaction of the metal with HCl and NaOH respectively. (c) What would happen if the metal is reacted with iron oxide?	$\frac{1}{2}+$ $2+$ $\frac{1}{2}=3$
28.	<u>*Attempt either option (a) or (b).*</u> (a) In the following sequence of reactions, identify X, Y, Z, P, Q, and B.  <u>**OR**</u> (b)(A) A milkman adds a very small amount of baking soda to fresh milk. (i) Why does he shift the pH of the fresh milk from 6 to slightly alkaline? (ii) Why does this milk take a long time to set as curd? (B) Why do acids not show acidic behaviour in the absence of water?	3
29.	Give Reasons: (a) Rings of cartilage are present in the trachea. (b) ATP is the energy currency for most cellular respiration. (c) Digestion cannot start without saliva.	3
30.	A cross was carried out between a pure breed tall pea plant and a pure breed dwarf pea plant and F_1 progeny was obtained. Later the F_1 progeny was served to obtained F_2 Progeny. Answer the followings questions: (a) What is the phenotype of the F_1 progeny and why? (b) Give the phenotypic ratio of the F_2 progeny. (c) Why is the F_2 progeny different from the F_1 progeny?	3
31.	(a) A lens of focal length 5 cm is being used by Debashree in the laboratory as a magnifying glass. Her least distance of distinct vision is 25 cm. (i) What is the	3

	magnification obtained by using the glass? (ii) She keeps a book at a distance 10 cm from her eyes and tries to read. She is unable to read. What is the reason for this? (b) Ravi kept a book at a distance of 10 cm from the eyes of his friend Hari. Hari is not able to read anything written in the book. Give reasons for this?	
32.	(i) State the law that explains the heating effect of current with respect to the measurable properties in an electrical circuit. (ii) List the factors on which the resistance of a conductor depends.	3
33.	Anannya responded to the question: Why do electrical appliances with metallic bodies are connected to the mains through a three-pin plug, whereas an electric bulb can be connected with a two-pin plug? She wrote: Three pin connections reduce heating of connecting wires. (i) Is her answer correct or incorrect? Justify. (ii) What is the function of a fuse in a domestic circuit?	3
SECTION D *This section comprises long answer (LA) type questions of 5 marks each.* **[3 x 5 = 15]**		
34.	*Attempt either option (a) or (b).* (a) (i) “Keerthi thinks that Substitution reaction occurs in saturated Hydrocarbons, on the contrary Krishi thinks, it occurs in unsaturated Hydrocarbons.” Justify with valid reasoning whose thinking is correct. (ii) “Methane and Propane and their Isomers are used as fuels” Comment. Draw the electron dot structure of the immediate lower homologue of Propane. Give any two characteristics of homologues of a given homologous series. (iii) A mixture of oxygen and ethyne is burnt for welding. Can you predict why a mixture of ethyne and air is not used? **OR** (b) (i) How do the following conversions take place? Write chemical equation for each A. Ethanol to ethanoic acid B. Ethene to ethane. (ii) Give one example each with chemical equation for the following reactions: A. Combustion reaction. B. Substitution reaction C. Saponification reaction.	1+3+1=5 **OR** 1x2+1x3=5
35.	(a)(i) Identify what is being shown in the figure. (ii) Name the part marked ‘A’ in the diagram. (iii) How does ‘A’ reach part ‘B” (iv) State the importance of part ‘C’. (v) What happens to the part marked ‘D’ after fertilization is over?  **OR** (b) i. Mention two functions of cotyledon. ii. How does the uterus prepare itself and nurture the growing embryo? What happens when the egg is not fertilized? iii. A woman is using a copper-T. Will it help in protecting her from sexually transmitted diseases?	5
36.	A person is unable to see objects distinctly placed within 75 cm from his eyes. (a) Name the defect of vision the person is suffering from. (b) List its two possible causes. (c) Calculate the power of the lens needed to correct this defect. Assume that the near point for the normal eye is 25 cm. **OR** (a) Why is a normal eye not able to see clearly the objects placed closer than 25 cm?	5 5

	(b) With the help of a diagram show recombination of the spectrum of white light. (c) List two essential conditions for observing a rainbow.	
	SECTION E *This section comprises 3 case study-based questions of 4 marks each.***[3 x 4 = 12]**	
	Case Study – 1	
37.	**<u>Read the passage carefully and answer the questions given below:</u>** Simmi is a Class X student residing in Delhi. After her exams, her family decided to take a trip to Mumbai. Simmi found that even after adding too much soap, she was unable to wash her clothes in Mumbai. Her mother suggested that she use detergent, as the water in Mumbai is hard. Simmi was surprised that clothes could be easily washed in Delhi with a small amount of soap, but in Mumbai, they could not be washed at all. (a) How soaps and detergents are different chemically? (b) Explain the formation of micelle in water. (c) Why detergents can easily work on hard water but soaps could not? **OR** Why soap solution appears cloudy?	1+2+1=4
	Case Study - 2	
38.	**<u>Read the passage carefully and answer the questions given below:</u>** Human body is made up of five important components, of which, water is the main component. Food as well as potable water are essential for every human being. The food is obtained from plants through agriculture. Pesticides are being used extensively for a high yield in the fields. These pesticides are absorbed by the plants from the soil along with water and minerals and from the water bodies these pesticides are taken up by the aquatic animals and plants. As these chemicals are not biodegradable, they get accumulated progressively at each trophic level. The maximum concentration of these chemicals gets accumulated in our bodies and greatly affects the health of our mind and body. (a) Why is the maximum concentration of pesticides found in human beings? (b) Give one method which could be applied to reduce our intake of pesticides through food to some extent. (c) What does the various steps in a food chain represent? **OR** Define food web.	2+1+1= 4
	Case Study - 3	
39.	**<u>Read the passage carefully and answer the questions given below:</u>** Rahima was investigating a circuit for her school project. She wanted to demonstrate the effect of length on the resistance of a conductor. The following is the given circuit with resistors A and B that are made of the same metal and the same thickness but ‘A’ is twice as long as ‘B’. The total current in the circuit is 6 A and the Voltage of the battery is 12 V.	4



	a) What will be the resistance in the circuit? b) Determine the value of ‘R_A’ and ‘R_B’. c) Determine the current in both the ammeters. Will the current in ‘A_1 ‘and ‘A_2’ be the same? Justify your answer. **OR** c) Define resistivity. What are the factors affecting the resistivity of a conductor?	(1) (1) (2) (2)